

Mat 1575 HW2: § 1.1

1. State whether the given sums are equal or unequal.

a. $\sum_{i=1}^{10} i$ and $\sum_{k=1}^{10} k$ They are equal: $\sum_{i=1}^{10} i = 1+2+3+4+5+6+7+8+9+10 = \sum_{k=1}^{10} k$.

b. $\sum_{i=1}^{10} i$ and $\sum_{i=6}^{15} (i-5)$ $\sum_{i=1}^{10} i = 1+2+3+4+5+6+7+8+9+10$;

and $\sum_{i=6}^{15} (i-5) = (6-5) + (7-5) + (8-5) + (9-5) + (10-5) + (11-5) + (12-5) + (13-5) + (14-5) + (15-5)$
 $= 1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10$

$\Rightarrow$ They are equal.

c. $\sum_{i=1}^{10} i(i-1)$ and $\sum_{i=1}^{10} i(i-1) = \sum_{i=1}^{10} (i^2 - i) = \sum_{i=1}^{10} i^2 - \sum_{i=1}^{10} i = \frac{5 \cdot 11 \cdot 21}{6} - \frac{5 \cdot 11}{2} = 330$

$\sum_{j=0}^9 (j+1)j = \sum_{j=0}^9 (j^2 + j) = \sum_{j=0}^9 j^2 + \sum_{j=0}^9 j = \frac{3 \cdot 10 \cdot 19}{6} + \frac{9 \cdot 10}{2} = 330$

$\Rightarrow$ They are equal

d. $\sum_{i=1}^{10} i(i-1)$ and $\sum_{i=1}^{10} i(i-1) = \sum_{i=1}^{10} i^2 - i$ which is equal to $\sum_{k=1}^{10} k^2 - k$

In the following exercises, use the rules for sums of powers of integers to compute the sums.

3. $\sum_{i=5}^{10} i^2 = \sum_{i=1}^{10} i^2 - \sum_{i=1}^4 i^2 = \frac{5 \cdot 11 \cdot 21}{6} - \frac{4 \cdot 5 \cdot 9}{6} = 385 - 30 = 355$

Suppose that $\sum_{i=1}^{100} a_i = 15$ and $\sum_{i=1}^{100} b_i = -12$. In the following exercises, compute the sums.

5. $\sum_{i=1}^{100} (a_i - b_i) = \sum_{i=1}^{100} a_i - \sum_{i=1}^{100} b_i = 15 - (-12) = 27$

7. $\sum_{i=1}^{100} (5a_i + 4b_i) = 5 \left(\sum_{i=1}^{100} a_i \right) + 4 \left(\sum_{i=1}^{100} b_i \right) = 5(15) + 4(-12) = 75 - 48 = 27$

Let L_n denote the left-endpoint sum using n subintervals and let R_n denote the corresponding right-endpoint sum. In the following exercises, compute the indicated left and right sums for the given functions on the indicated interval.

12. L_4 for $f(x) = \frac{1}{x-1}$ on $[2, 3]$: $n=4$ means there is a partition P such that it cuts $[2, 3]$ into 4 pieces evenly with $\Delta x = \frac{3-2}{4} = \frac{1}{4}$ and,

$$P = \{2, 2+\Delta x, 2+2\Delta x, 2+3\Delta x, 2+4\Delta x\} = \{2, \frac{9}{4}, \frac{10}{4}, \frac{11}{4}, 3\}.$$

$$\begin{aligned} \text{Then } L_4 &= f(2) \cdot \Delta x + f(\frac{9}{4}) \Delta x + f(\frac{10}{4}) \Delta x + f(\frac{11}{4}) \Delta x \\ &= (\frac{1}{2-1} + \frac{1}{\frac{9}{4}-1} + \frac{1}{\frac{10}{4}-1} + \frac{1}{\frac{11}{4}-1}) \cdot \Delta x \\ &= (1 + \frac{4}{5} + \frac{4}{6} + \frac{4}{7}) \cdot \frac{1}{4} = (\frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7}) = \frac{209}{420} \end{aligned}$$

15. R_6 for $f(x) = \frac{1}{x(x-1)}$ on $[2, 5]$ $n=6$ means there is a partition P such that it cuts $[2, 5]$ into 6 pieces evenly $\Delta x = \frac{5-2}{6} = \frac{3}{6} = \frac{1}{2}$:

$$P = \{2, 2+\Delta x, 2+2\Delta x, 2+3\Delta x, 2+4\Delta x, 2+5\Delta x, 2+6\Delta x\}$$

$$= \{2, \frac{5}{2}, 3, \frac{7}{2}, 4, \frac{9}{2}, 5\} \text{ Then}$$

$$\begin{aligned} R_6 &= f(\frac{5}{2}) \Delta x + f(3) \Delta x + f(\frac{7}{2}) \Delta x + f(4) \Delta x + f(\frac{9}{2}) \Delta x + f(5) \Delta x \\ &= (\frac{1}{\frac{5}{2}-1} - \frac{1}{\frac{5}{2}} + \frac{1}{3-1} - \frac{1}{3} + \frac{1}{\frac{7}{2}-1} - \frac{1}{\frac{7}{2}} + \frac{1}{4-1} - \frac{1}{4} + \frac{1}{\frac{9}{2}-1} - \frac{1}{\frac{9}{2}} + \frac{1}{5-1} - \frac{1}{5}) \Delta x \\ &= (\frac{2}{3} - \frac{2}{5} + \frac{1}{2} - \frac{1}{3} + \frac{2}{5} - \frac{2}{7} + \frac{1}{3} - \frac{1}{4} + \frac{2}{7} - \frac{2}{9} + \frac{1}{4} - \frac{1}{5}) \cdot \frac{1}{2} \\ &= (\frac{2}{3} + \frac{1}{2} - \frac{2}{9} - \frac{1}{5}) \frac{1}{2} = (\frac{7}{6} - \frac{19}{45}) \frac{1}{2} = \frac{105-38}{90} \cdot \frac{1}{2} = \frac{67}{90} \cdot \frac{1}{2} = \frac{67}{180} \end{aligned}$$

16. R_4 for $\frac{1}{x^2+1}$ on $[-2, 2]$ $n=4$ means we have $\Delta x = \frac{2-(-2)}{4} = \frac{4}{4} = 1$ and

$$P = \{-2, -2+\Delta x, -2+2\Delta x, -2+3\Delta x, -2+4\Delta x\}$$

$$= \{-2, -1, 0, 1, 2\} \text{ Then}$$

$$\begin{aligned} R_4 &= f(-1) \Delta x + f(0) \Delta x + f(1) \Delta x + f(2) \Delta x \\ &= (\frac{1}{(-1)^2+1} + \frac{1}{(0)^2+1} + \frac{1}{(1)^2+1} + \frac{1}{(2)^2+1}) \Delta x \end{aligned}$$

$$= \left(\frac{1}{2} + 1 + \frac{1}{2} + \frac{1}{5} \right) \cdot 1 = \frac{11}{5}$$

17. L_4 for $\frac{1}{x^2+1}$ on $[-2, 2]$ Similarly as in 16, $n=4$, $\Delta x = \frac{2-(-2)}{4} = 1$ and

$$P = \{-2, -1, 0, 1, 2\}$$

$$L_4 = f(-2)\Delta x + f(-1)\Delta x + f(0)\Delta x + f(1)\Delta x$$

$$= \left(\frac{1}{(-2)^2+1} + \frac{1}{(-1)^2+1} + \frac{1}{(0)^2+1} + \frac{1}{(1)^2+1} \right) \Delta x$$

$$= \left(\frac{1}{5} + \frac{1}{2} + 1 + \frac{1}{2} \right) \cdot 1 = \frac{11}{5}$$